Groups: Basic Properties

From the class book: Chapter 4, Sections 4.1 to 4.4

1. Key definitions

Binary operation. A map $* : G \times G \to G$ on a non-empty set G. On an n-element set there are n^{n^2} binary operations, of which $n^{n(n+1)/2}$ are commutative.

Let $*$ be a binary operation on G. Then $(G, *)$ is

- a **groupoid** (magma): just a set with a binary operation;
- a **semigroup** if $*$ is associative;
- a **monoid** if it is a semigroup with an identity e ($ae = ea = a$);
- a **group** if it is a monoid in which every a has an inverse a^{-1} ($aa^{-1} = a^{-1}a = e$);
- an **abelian group** if moreover $ab = ba$ for all a, b.

Order of a group. $|G|$ or $o(G)$ is the number of elements of G.

Order of an element. $o(a)$ is the least $n \geq 1$ with $a^n = e$; if there is none, $o(a) = \infty$. Equivalently, $o(a) = |\langle a \rangle|$.

Which structure is it?

Structure	Associative	Identity	Inverses	Type
$(\mathbb{Z}, -)$	no	no	n/a	groupoid only
$(\mathbb{N}, +)$	yes	no	no	semigroup
$(\mathbb{N}, \cdot)$, $(\mathbb{Z}, \cdot)$	yes	1	no	monoid
$(P(X), \cup)$, $(P(X), \cap)$	yes	$\emptyset$, X	no	monoid
$(\mathbb{N}, \mathrm{lcm})$	yes	1	no	monoid
$(\mathbb{N}, \gcd)$	yes	no	no	semigroup
$(\mathbb{N}, a * b = a^b)$	no	no	n/a	groupoid only
$(\mathbb{Z}, +)$, $(P(X), \triangle)$	yes	0, $\emptyset$	yes	abelian group

2. Main results

Theorem (When is a semigroup a group?)
A semigroup G is a group iff the equations $ax = b$ and $ya = b$ are solvable in G for all a, b. A *finite* semigroup is a group iff both cancellation laws hold.

Elementary properties. In every group the identity and inverses are unique, both cancellation laws hold, $ax = b$ has the unique solution $x = a^{-1}b$ (and $ya = b$ has $y = ba^{-1}$), and $(ab)^{-1} = b^{-1}a^{-1}$.

Theorem (Criteria for a group to be abelian)

Each of the following implies that G is abelian:

- $a^2 = e$ for every $a \in G$;
- $(ab)^2 = a^2 b^2$ for all a, b;
- $(ab)^{-1} = a^{-1} b^{-1}$ for all a, b;
- $x \mapsto x^{-1}$ (or $x \mapsto x^2$) is a homomorphism;
- $|G| \le 5$, or $|G| = p$ or p^2 with p prime.

The smallest non-abelian group is S_3, of order 6.

Proof idea for the second: $abab = aabb$; cancel a on the left and b on the right to get $ba = ab$. For the first: $ab = (ab)^{-1} = b^{-1}a^{-1} = ba$.

Theorem (Properties of the order)

Let $o(a) = n < \infty$. Then

- $a^m = e \iff n \mid m$;
- $o(a^k) = \dfrac{n}{\gcd(n, k)}$;
- $o(a) = o(a^{-1}) = o(xax^{-1})$ and $o(ab) = o(ba)$;
- if $ab = ba$ and $\gcd(o(a), o(b)) = 1$, then $o(ab) = o(a)\,o(b)$;
- in a finite group, $o(a)$ divides $|G|$ (Lagrange).

Counting elements of a given order

- The only element of order 1 is e.
- Elements of order $k \ge 3$ come in pairs $\{a, a^{-1}\}$, so their number is even; in fact it is a multiple of $\varphi(k)$.
- A group of even order has an odd number of elements of order 2 (so at least one); a group of odd order has none.
- (Frobenius) If $k \mid |G|$, the number of solutions of $x^k = e$ in G is a multiple of k.
- If G has a unique element of order 2, it lies in the centre $Z(G)$.
- An infinite group can have all elements of finite order: $(P(\mathbb{N}), \triangle)$, $\mathbb{Q}/\mathbb{Z}$.

Theorem (Orders of linear groups)

$$|GL(n, \mathbb{F}_q)| = (q^n - 1)(q^n - q)\cdots(q^n - q^{n-1}), \qquad |SL(n, \mathbb{F}_q)| = \frac{|GL(n, \mathbb{F}_q)|}{q - 1}.$$

Proof idea: choose the rows of an invertible matrix one at a time; the k-th row must avoid the q^{k-1} vectors spanned by the earlier rows. The determinant map onto $\mathbb{F}_q^*$ has kernel SL.

3. Standard groups

Group	Order	Abelian?	Remarks
$(\mathbb{Z},+),(\mathbb{Q},+),(\mathbb{R},+),(\mathbb{C},+)$	infinite	yes	$\mathbb{Z}$ cyclic; $\mathbb{Q},\mathbb{R},\mathbb{C}$ not cyclic
$\mathbb{Q}^*,\mathbb{R}^*,\mathbb{C}^*$ (multiplication)	infinite	yes	$\mathbb{R}^*$ has only ± 1 of finite order
$(\mathbb{Z}_n,+)$	n	yes	cyclic, generated by $\bar{1}$
$U(n)=\{\bar{a} : \gcd(a,n)=1\}$, multiplication	$\varphi(n)$	yes	cyclic iff $n=1,2,4,p^k,2p^k$ (p odd prime)
$\mu_n=\{z\in\mathbb{C} : z^n=1\}$	n	yes	cyclic, generator $e^{2\pi i/n}$
$K_4=\{e,a,b,ab\}$	4	yes	Klein group, $\cong U(8)\cong\mathbb{Z}_2\times\mathbb{Z}_2$
$Q_8=\{\pm 1,\pm i,\pm j,\pm k\}$	8	no	quaternion group
D_n (symmetries of an n-gon)	$2n$	no ($n\geq 3$)	dihedral group
S_n	$n!$	no ($n\geq 3$)	symmetric group
$GL(n,F)$, $SL(n,F)$		no ($n\geq 2$)	general and special linear groups
$(P(X),\triangle)$	$2^{\lvert X\rvert}$	yes	every element is its own inverse

The quaternion group

$Q_8=\{\pm 1,\pm i,\pm j,\pm k\}$ with $i^2=j^2=k^2=-1$, $ij=k=-ji$, $jk=i=-kj$, $ki=j=-ik$. Presentation: $\langle a,b \mid a^4=e,\ a^2=b^2,\ b^{-1}ab=a^{-1}\rangle$. Matrix model:

$$i=\begin{pmatrix} i & 0\\ 0 & -i\end{pmatrix},\quad j=\begin{pmatrix} 0 & 1\\ -1 & 0\end{pmatrix},\quad k=\begin{pmatrix} 0 & i\\ i & 0\end{pmatrix}.$$

Element orders: one of order 1, one of order 2 (namely -1), six of order 4.

The dihedral group

$D_n=\langle r,s \mid r^n=s^2=e,\ srs^{-1}=r^{-1}\rangle=\{e,r,\dots,r^{n-1},s,sr,\dots,sr^{n-1}\}$: n rotations and n reflections. Each reflection sr^k has order 2; the rotation r^k has order $n/\gcd(k,n)$. Elements of order 2: n if n is odd, $n+1$ if n is even (the extra one is $r^{n/2}$).

Linear groups over finite fields

Group	$GL(2,\mathbb{Z}_2)$	$GL(2,\mathbb{Z}_3)$	$SL(2,\mathbb{Z}_3)$	$GL(2,\mathbb{Z}_5)$	$GL(3,\mathbb{Z}_2)$
Order	6 ($\cong S_3$)	48	24	480	168

In general $\lvert GL(2,\mathbb{Z}_p)\rvert=(p^2-1)(p^2-p)$ and $\lvert SL(2,\mathbb{Z}_p)\rvert=p(p^2-1)$.

How to find orders

- In $\mathbb{Z}_n$: $o(\bar{a})=n/\gcd(a,n)$.
- In S_n: the lcm of the cycle lengths.
- In $G\times H$: $o(a,b)=\operatorname{lcm}(o(a),o(b))$.
- For a matrix: compute powers until I appears (a diagonal matrix has order the lcm of the orders of its entries).

4. Exam shortcuts and standard counter-examples

> **Exam shortcut.** Testing whether $(G,*)$ is a group: check closure, then associativity (the standard non-associative operations are $a-b$, a^b, $\frac{a+b}{2}$, $\lvert a-b\rvert$), then solve $a*e=a$ for an e independent of a, then solve $a*a'=e$. For a finite set, every row of the Cayley table must be a permutation of G.

Exam shortcut. For $\left\{\begin{pmatrix} x & x \\ x & x \end{pmatrix} : x \neq 0\right\}$ under matrix multiplication (a group, although these matrices are singular), the identity is $x = \frac{1}{2}$ and the inverse of x is $\frac{1}{4x}$. For $a * b = a + b + ab$ on $\mathbb{R} \setminus \{-1\}$, $e = 0$ and $a^{-1} = -\dfrac{a}{1+a}$.

Exam shortcut. $(\mathbb{Z}_n, \cdot)$ is never a group for $n \geq 2$, since $\bar{0}$ has no inverse; $(U(n), \cdot)$ always is.

Exam shortcut. Without commutativity $o(ab)$ can be anything. In the group of bijections of $\mathbb{R}$, $f(x) = -x$ and $g(x) = 1 - x$ both have order 2, but $g \circ f(x) = 1 + x$ has infinite order. Even for commuting elements, $o(ab) = o(a)o(b)$ needs $\gcd(o(a), o(b)) = 1$: take $b = a^{-1}$.

Exam shortcut. $a^2 = e$ for all a forces G abelian, but $a^3 = e$ for all a does not (the Heisenberg group of order 27 is a counter-example).

5. Solved examples

Example 1. On $G = \mathbb{R} \setminus \{-1\}$ define $a * b = a + b + ab$. Show that $(G, *)$ is an abelian group and find the inverse of 3.

Solution. Note $1 + a * b = (1 + a)(1 + b)$. So if $a, b \neq -1$ then $a * b \neq -1$ (closure), and $*$ is associative and commutative because multiplication is. The identity:
$a + e + ae = a \Rightarrow e(1 + a) = 0 \Rightarrow e = 0$. Inverse: $a + a' + aa' = 0 \Rightarrow a' = -\frac{a}{1+a}$, which is never -1. For $a = 3$: $a' = -\frac{3}{4}$. Check: $3 - \frac{3}{4} - \frac{9}{4} = 0$.

Example 2. Find the orders of all elements of $U(20)$. Is $U(20)$ cyclic?

Solution. $U(20) = \{1, 3, 7, 9, 11, 13, 17, 19\}$, of order $\varphi(20) = 8$. Squares mod 20:
$3^2 = 9,\ 7^2 = 49 \equiv 9,\ 9^2 = 81 \equiv 1,\ 11^2 = 121 \equiv 1,\ 13^2 = 169 \equiv 9,\ 17^2 = 289 \equiv 9,\ 19^2 = 361 \equiv 1$. Hence $9, 11, 19$ have order 2 and $3, 7, 13, 17$ have order 4 (their squares are $9 \neq 1$). No element has order 8, so $U(20)$ is not cyclic (consistent with $20 \neq 1, 2, 4, p^k, 2p^k$); in fact $U(20) \cong \mathbb{Z}_2 \times \mathbb{Z}_4$.

Example 3. Find (a) the number of elements of order 2 in D_8 (order 16), (b) the number of elements of order 4 in D_8, (c) the number of elements of order 2 in D_5.

Solution. (a) $n = 8$ is even: 8 reflections plus r^4, so 9.
(b) Reflections have order 2, so only rotations:
$o(r^k) = 8/\gcd(k, 8) = 4 \iff \gcd(k, 8) = 2 \iff k = 2, 6$. Answer 2 ($= \varphi(4)$).
(c) $n = 5$ is odd: only the 5 reflections. Answer 5 (odd, as it must be in a group of even order).

Example 4. Find $|GL(2, \mathbb{Z}_5)|$, $|SL(2, \mathbb{Z}_5)|$, and the orders of $A = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$ and $B = \begin{pmatrix} 2 & 0 \\ 0 & 3 \end{pmatrix}$ in $GL(2, \mathbb{Z}_7)$.

Solution. $|GL(2, \mathbb{Z}_5)| = (25 - 1)(25 - 5) = 24 \cdot 20 = 480$, and
$|SL(2, \mathbb{Z}_5)| = 480/4 = 120 = 5(25 - 1)$.
$A^m = \begin{pmatrix} 1 & m \\ 0 & 1 \end{pmatrix}$, which is I iff $7 \mid m$; so $o(A) = 7$.
$B^m = \operatorname{diag}(2^m, 3^m)$. Mod 7, $o(2) = 3$ ($2^3 = 8 \equiv 1$) and $o(3) = 6$ (3 is a primitive root mod 7). So $o(B) = \operatorname{lcm}(3, 6) = 6$.

Example 5. Let $o(a) = 24$. Find $o(a^{18})$, $o(a^{-10})$, and the number of k, $1 \le k \le 24$, with $o(a^k) = 24$. If b commutes with a and $o(b) = 5$, find $o(ab)$.

Solution. $o(a^{18}) = 24/\gcd(24, 18) = 24/6 = 4$. $o(a^{-10}) = o(a^{10}) = 24/2 = 12$.
$o(a^k) = 24 \iff \gcd(k, 24) = 1$: $\varphi(24) = 8$ values. Since $ab = ba$ and $\gcd(24, 5) = 1$, $o(ab) = 24 \cdot 5 = 120$.

Example 6. Let G be a group. Which of the following force G to be abelian? (a) $(ab)^{-1} = a^{-1}b^{-1}$ for all a, b; (b) $|G| = 49$; (c) $|G| = 8$; (d) $x^3 = e$ for all x.

Solution. (a) Yes: $b^{-1}a^{-1} = a^{-1}b^{-1}$ for all a, b; replacing a, b by their inverses gives $ba = ab$.
(b) Yes: $49 = 7^2$, and groups of order p^2 are abelian.
(c) No: Q_8 and D_4 have order 8 and are non-abelian.
(d) No: the non-abelian group of order 27 and exponent 3 (upper unitriangular 3×3 matrices over $\mathbb{Z}_3$).
Answer: (a), (b).